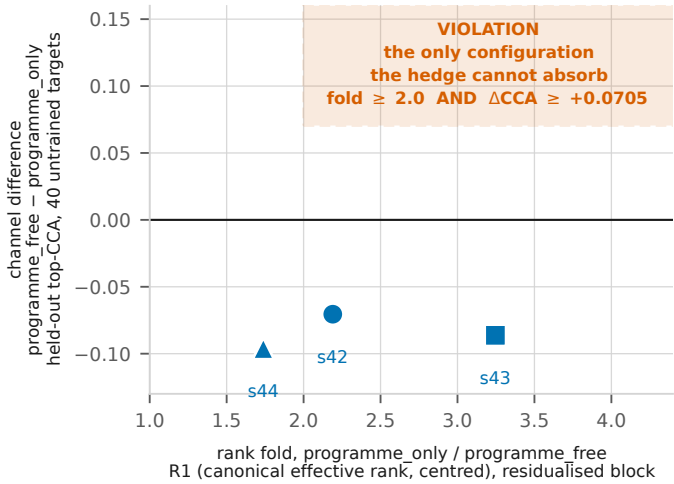
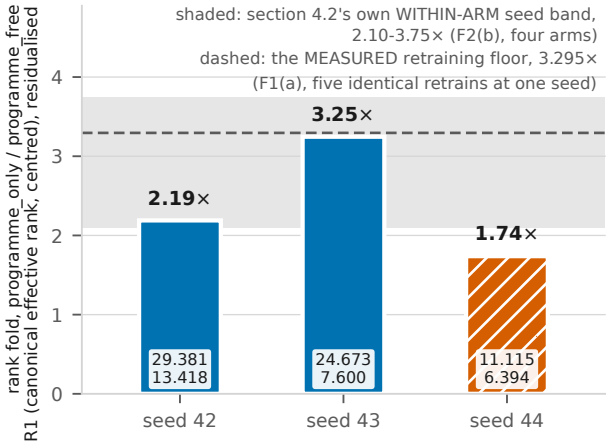


(a) The predeclaration, drawn



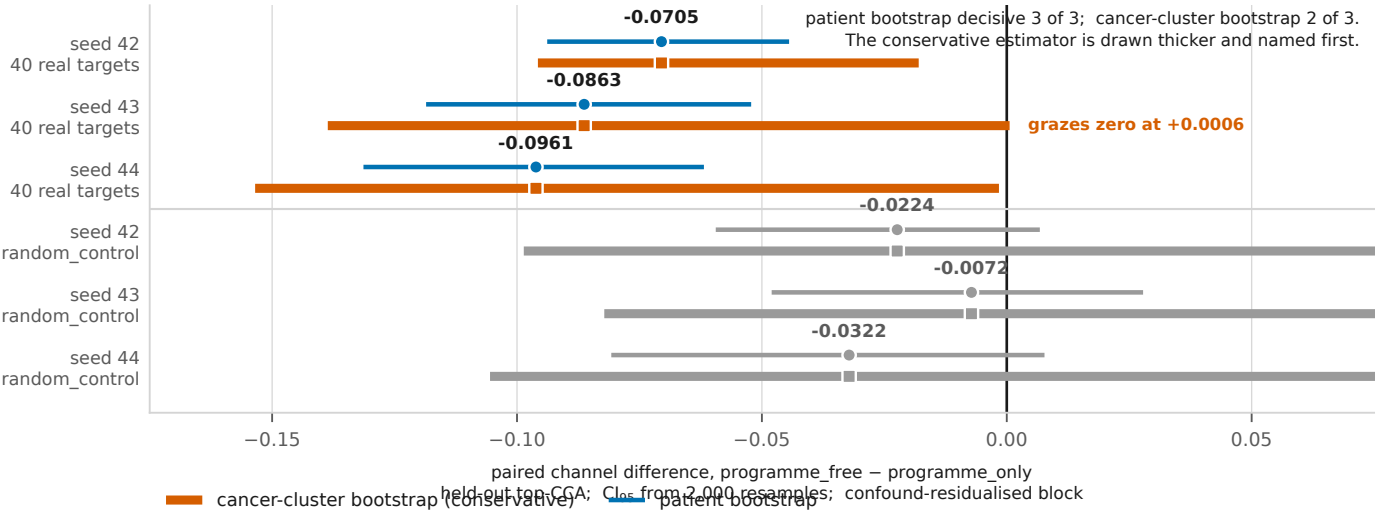
All three land in the CONFIRMING quadrant: lower rank, LESS information.
Both thresholds, and outcomes O1-O4, were pre-declared: see the caption.

(b) The three seeds, in rank



2 of 3 sit inside the nuisance band and 3 of 3 are below the MEASURED retraining floor, so D1 does not resolve rank in EITHER direction. That does not refute the necessity result opposite: the channel difference is resolvable where the rank difference is not. Bar labels are the two R1 values.

(b) ... and in the information channel, with BOTH bootstrap estimators - and (c) the negative control on the same axes



(d) The one violation, small and last

	lower-rank	rank	CCA	higher-rank	rank	CCA	fold	dCCA
P44		11.115	0.6087	I43	34.117	0.4882	3.07x	+0.1206
	cross-experiment							
H44		9.143	0.5983	I43	34.117	0.4882	3.73x	+0.1101
	within-experiment, across arms							

2 of 66 ordered pairs. Drawn deliberately small: both are cross-arm, one is cross-experiment, and RankMe reserves itself to "different runs of a given method". Partially pre-empted by Aldeneh et al. (ICASSP 2025), who published that lower-ranked layers can outperform higher-ranked ones.

THE PREREGISTERED ESCALATION THIS FIGURE TRIPS

D1_PAIR_MANIFEST.json records, before the run:
"programme_free >= programme_only on the held-out molecular channel; if programme_only wins, the collapse story is wrong -- escalate, do not proceed to D2"

programme_only won on all three seeds. This paper flags that and does not resolve it.

The arms differ in OBJECTIVE, so D1 sits closer to a between-method comparison than D2 and lands further outside RankMe's stated scope. Section 4.7 is therefore a CONFIRMATION of RankMe, not a refutation of it, and this figure carries the paper's own negative.

F6. This is the paper's own negative and it is reported before the instances that favour the paper. D1_PAISED_BOOTSTRAP.json (unsuffixed) MUST NOT be used and is not vendored here: it scores all 90 non-control targets, of which 50 are programme_only's own supervision. The intervals drawn are the 40-target STRATIFIED bootstrap and the 90-target RANDOM_CONTROL bootstrap. Both estimators are drawn and the conservative one is weighted: the patient bootstrap is decisive 3 of 3, the cancer-cluster bootstrap 2 of 3, and seed 43's cancer interval grazes zero - the panel shows that it does. The arms are matched on everything but the objective (D1_PAIR_MANIFEST.json, "objective_only_difference": true). Statistic R1, block confound-residualised, on the held-out wsi_biology block. PANEL (a)'S PREDECLARATION: the violation region is fold ≥ 2.0 AND dCCA $\geq +0.0705$, both fixed before the pair list was inspected. They are declared in v2/research/rebase/p2/p2_necessity_and_variance.py and quoted as pre-fixed in NOTEBOOK_ENTRIES/p2_competing_metrics_and_necessity_test_20260803T2326Z.md; the four outcomes O1-O4 and how each would be read were pre-declared 2026-08-03 23:00 UTC in NOTEBOOK_ENTRIES/PREDECLARED_D1_necessity_test_20260803T2300Z.md, before any channel value had been seen.